

# Investigation 19: Conservation of mass

## Notes

An understanding of the concept of conservation of mass is fundamental to chemistry. During a chemical reaction atoms are not created or destroyed, but are rearranged. As a result of this, the total mass of the reactants consumed will always equal the total mass of the products formed.

### The task

Design an experiment that verifies the Law of Conservation of Mass using the reaction between sodium hydrogencarbonate and hydrochloric acid.

### Equipment

sodium hydrogencarbonate  
hydrochloric acid solution  $2 \text{ mol L}^{-1}$   
other equipment as specified by your investigation design

### Planning the investigation

1. Plan an investigation to demonstrate the Law of Conservation of Mass.
2. Write out a proposed procedure for the investigation including any equipment and chemicals you require. Identify any safety requirements.
3. Check the proposed procedure with your teacher

### Conducting the investigation

1. Conduct the investigation.
2. If time allows replicate the experiment.

### Processing the data

Does your investigation confirm the law of conservation of mass? Justify your answer by referring to the data you have collected.

### Evaluating the investigation

Evaluate the effectiveness of your procedure and describe any modifications you would make to improve the accuracy of the results and your organisation of the investigation. You should consider the following:

- What other reactions could be used to verify the law of conservation of mass?
- What problems, if any, were there in using a reaction that produces a gas?